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United States Patent	12397304
Kind Code	B2
Date of Patent	August 26, 2025
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Dry fire protection sprinkler and fluid deflection member assemblies

Abstract

A dry fire protection sprinkler assembly. The sprinkler assembly includes an elongate tubular outer housing having a first end and a second end opposite the first end. The second end of the housing defines a fluid discharge end of the sprinkler assembly having an outlet opening. A fluid deflection member is affixed to the housing at a preferably fixed distance from the outlet opening. The sprinkler assembly and the fluid deflection member define an unencumbered fluid flow path for a column of fluid discharge from the outlet opening. The fluid deflection member includes at least one tab to act on the unencumbered column of fluid discharge.

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Appl. No.:	17/912927
Filed (or PCT Filed):	March 30, 2021
PCT No.:	PCT/US2021/024905
PCT Pub. No.:	WO2021/202543
PCT Pub. Date:	October 07, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230149953 A1	May. 18, 2023

Related U.S. Application Data

Publication Classification**Int. Cl.:** B05B1/26 (20060101); A62C37/14 (20060101)**U.S. Cl.:****CPC** B05B1/267 (20130101); A62C37/14 (20130101);**Field of Classification Search****CPC:** A62C (35/62); A62C (37/11); A62C (37/12); A62C (37/14); A62C (37/16); B05B (1/262); B05B (1/267)

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Background/Summary

PRIORITY DATA AND INCORPORATION BY REFERENCE (1) This application is a 35 U.S.C. § 371 application of International Application No. PCT/US2021/024905, filed Mar. 30, 2021, which claims the benefit of U.S. Provisional Application No. 63/003,660 filed Apr. 1, 2020, each of which is incorporated by reference in its entirety.

TECHNICAL FIELD

(1) The present invention relates generally to dry sprinkler assemblies and in particular, dry fire protection sprinkler and fluid deflection member assemblies.

BACKGROUND ART

(2) Generally, automatic fire protection sprinklers include a sprinkler frame and/or housing having an inlet, an outlet and internal passageway through which firefighting fluid flows and discharged to impact a fluid deflection member that is coupled to the sprinkler frame and spaced from the outlet. Fluid flow through the sprinkler is controlled by a thermally responsive trigger which supports a sealing assembly in a position that seals the internal passageway of the sprinkler. The trigger has a nominal operating temperature and thermal sensitivity to define the thermal responsiveness of the sprinkler at which the sprinkler actuates in response to a fire. Upon thermal actuation of the trigger in response to a fire, the trigger fractures or collapses thereby releasing the sealing assembly to allow the flow of fluid through the sprinkler internal passageway, out the outlet and toward the fluid deflection member. Fluid deflection members can be formed to a variety of geometries to suit a given fire protection application. The deflector geometries can be categorized into one of two types. One type of fluid deflection member presents a central abutment to the fluid discharge from the outlet opening and fans the fluid discharge radially. Such a deflector geometry is shown, for example, in U.S. Pat. Nos. 7,766,252 and 5,664,630.

(3) An alternate type of deflection geometry defines an unencumbered fluid flow path. As used herein, an “unencumbered fluid flow path” provides for a fluid discharge column in which its central core is not impacted by any sprinkler structure and fanned radially. Instead, the fluid deflection member geometry acts on the periphery of the discharge column to direct the fluid stream in a desired manner. Such a deflector geometry is shown, for example, in the flow-shaper members of U.S. Pat. Nos. 8,662,190; 8,151,462 and 7,712,218. In each of the sprinkler assemblies shown, the fluid deflection member is located and affixed downstream of the sprinkler outlet with the sprinkler trigger seated against a yoke member to support the seal assembly in the sprinkler passageway. Each of the seal assembly, trigger and yoke member are located between the outlet and the fluid deflection member. In order to ensure that the central core of fluid discharge is not impacted by the trigger or supporting structures upon thermal actuation of the trigger, the sealing assembly includes a sealing spring that acts against the sealing surface to bias or urge the sealing assembly out of the outlet. Moreover, the yoke member is structured to increase its instability in the assembly upon trigger actuation so that the yoke member falls out of the fluid flow path along with the ejected sealing assembly. In the sprinkler assemblies of U.S. Pat. Nos. 8,662,190; 8,151,462 and 7,712,218, the seal assembly forms a fluid tight seal against an internal sealing surface located

at or proximate the outlet. Accordingly, firefighting fluid delivered to such an installed sprinkler assembly, fills the sprinkler up to the seal assembly proximate the outlet. The presence of the fluid proximate the outlet provides fluid pressure to act against the seal assembly, trigger and yoke member to clear them from the fluid flow path upon trigger actuation and maintain the unencumbered fluid flow path.

(4) Another type of automatic sprinkler is the dry sprinkler assembly. An example of a dry sprinkler is shown in U.S. Pat. No. 8,636,075. Dry sprinklers can be configured for installation in a variety of orientations depending upon the application. Dry sprinklers can be configured for an upright installation, a pendent installation or a horizontal installation. An example of a pendent-type dry sprinkler is shown and described in U.S. Pat. No. 5,664,630. An example of a horizontal dry sprinkler is shown and described in U.S. Pat. No. 7,921,928. A dry sprinkler assembly generally includes a tubular sprinkler housing with an inlet end fluid opening and a discharge outlet opening axially spaced from the inlet opening with an internal passageway extending therebetween. An internal fluid control assembly is supported within the housing between the inlet and outlet openings by a frangible thermally responsive glass bulb trigger to seal the sprinkler at the fluid inlet. When the bulb fractures in response to a fire, a component of the fluid control assembly is ejected from the outlet of the housing allowing the remainder of the fluid control assembly to axially translate out of its sealed position thereby opening the fluid inlet and sprinkler internal passageway. To ensure proper opening and operation of a dry sprinkler assembly, it is important that the ejected member completely clear the sprinkler structure and fluid flow path between the housing and the fluid deflection member. The sprinkler shown and described in U.S. Pat. No. 7,921,928 includes a sprinkler body attached to the housing to define the outlet opening. As shown, the sprinkler body includes a pair of frame arms that extend away from the outlet opening and converge toward a coaxially centrally aligned fluid deflection boss, to which the fluid deflection member is affixed. The fluid deflection boss and the fluid deflection member present a central abutment to the fluid discharge from the sprinkler outlet opening to redirect and spread the discharged fluid from its center and provide an effective horizontal fluid distribution. It is believed that, there remains a need for alternate dry sprinkler and fluid deflection member assemblies, and in particular, for dry sprinkler assemblies with fluid deflection members that provide for an unencumbered fluid flow path.

DISCLOSURE OF INVENTION

(5) Preferred embodiments of a dry fire protection sprinkler and fluid deflection member assembly are provided. The preferred sprinkler assembly generally includes an elongate tubular outer housing having a first end and a second end opposite the first end. Within the tubular housing, an internal conduit extends from the first end to the second end along a longitudinal sprinkler axis. The first end of the housing defines a fluid intake end of the sprinkler assembly having an inlet opening and an internal sealing surface proximate the inlet opening. The second end of the housing defines a fluid discharge end of the sprinkler assembly having an outlet opening. A fluid deflection member is affixed to the housing at a preferably fixed distance from the outlet opening to define an unencumbered fluid flow path.

(6) The sprinkler assembly is preferably an automatic sprinkler in which fluid flow through the sprinkler is regulated by a thermally responsive trigger assembly and a preferred internal fluid control assembly disposed within the housing. The trigger defines an unactuated state of the sprinkler assembly in which the trigger supports the internal fluid control assembly within the housing to form a fluid tight seal with the internal sealing surface. Upon thermal operation of the trigger, an actuated state of the sprinkler assembly is defined in which the internal fluid control assembly is displaced and preferably axially translated out of contact with the internal sealing surface. Preferred embodiments of the fluid control assembly include a support subassembly that provide an ejectable member that is ejected out of the housing upon sprinkler actuation. Upon sprinkler actuation, preferred embodiments of the ejectable member of the fluid control assembly

define a mechanical interface with other structures of the sprinkler assembly which facilitates ejection of the ejectable member through the housing outlet opening. Moreover, the preferred mechanical interface facilitates clearance of the ejectable member and other sprinkler components out of the preferred unencumbered fluid flow path of the sprinkler assembly.

(7) Preferred embodiments of a dry sprinkler assembly include a tubular outer housing having a first end and a second end opposite the first end with an internal conduit extending from the first end to the second end along a central longitudinal sprinkler axis. The first end of the housing defines a fluid intake end of the sprinkler assembly having an inlet opening and an internal sealing surface proximate the inlet opening. The second end of the housing defines a fluid discharge end of the housing having an outlet opening. The second end of the housing also preferably includes a terminal portion of the housing that circumscribes about the central longitudinal axis. The preferred terminal portion of the housing is axially spaced from the outlet opening to preferably define a frame window between the terminal portion and the outlet opening. A fluid control assembly is disposed coaxially within the internal conduit of the outer housing for displacement from being in fluid tight contact with the sealing surface to being out of contact with the sealing surface to provide for an unencumbered fluid flow path extending along the central longitudinal sprinkler axis from the fluid control assembly through the terminal portion. A preferred fluid deflection member is affixed to the terminal portion of the housing. The fluid deflection member preferably includes a fluid flow surface to maintain the unencumbered fluid flow path along the central longitudinal sprinkler axis between the fluid control assembly to the fluid flow surface when the fluid control assembly is out of contact with the sealing surface. Preferred embodiments of the fluid deflection member include at least one tab radially spaced from the central longitudinal axis at a fixed distance from the outlet opening to define the preferred fluid flow surface and maintain the unencumbered fluid flow path along the central longitudinal sprinkler axis extending between the fluid control assembly and the at least one tab.

(8) In another preferred embodiment of the sprinkler assembly, the tubular outer housing includes a pair of frame arms diametrically opposed about the outlet opening extending axially therefrom to define a frame window therebetween. A preferred fluid deflection member is coupled to the frame arms at the preferred fixed distance from the outlet opening. Preferred embodiments of the sprinkler assembly and its fluid deflection member define an unencumbered fluid flow path for a column of fluid discharge from the outlet opening of the housing. In one preferred embodiment, the fluid deflection member includes at least one tab, and more preferably includes at least two tabs opposed from one another about a plane that includes the central longitudinal sprinkler axis. The preferred tabs act on an unencumbered column of fluid discharge from the outlet opening in a radially inward direction. In preferred embodiments, each of the tabs are symmetrical about a second plane that is perpendicular to and intersects the first plane along the central longitudinal sprinkler axis with the pair of frame arms being aligned with one another in the second plane. The pair of frame arms preferably terminate at an annular boss centered about the central longitudinal sprinkler axis with the preferred fluid deflection member being affixed to the annular frame boss.

(9) Preferred embodiments of the sprinkler assembly include a trigger embodied as a frangible glass bulb having a first end seated against the support subassembly and an opposite second end seated against a yoke member to align the glass bulb along the central sprinkler axis. The yoke preferably includes a crossbar portion with a central region for seating the second end of the glass bulb and two end regions disposed about the central region that are each subject to a load force to axially load the glass bulb and fluid control assembly. The preferred yoke member includes an extension member extending between the two end regions of the yoke member to define a center of gravity that is off-set from the central longitudinal sprinkler axis.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) The accompanying drawings, which are incorporated herein and constitute part of this specification, illustrate exemplary embodiments of the invention, and together, with the general description given above and the detailed description given below, serve to explain the features of the invention. It should be understood that the preferred embodiments are some examples of the invention as provided by the appended claims.
- (2) FIGS. **1**, **1A** and **1B** are various cross-sectional views of a preferred embodiment of a dry sprinkler assembly.
- (3) FIG. **2** is a preferred fluid deflection member for use in the sprinkler assembly of FIG. **1**.
- (4) FIGS. **3A-3B** are various detailed partial cross-sectional views at the fluid discharge end of the sprinkler assembly of FIG. **1**.
- (5) FIG. **4** is a perspective view of a support subassembly used in the sprinkler assembly of FIG. **1**.
- (6) FIG. **4A** is an exploded perspective view of the support subassembly of FIG. **4**.

MODE(S) FOR CARRYING OUT THE INVENTION

- (7) Shown in FIGS. **1**, **1A** and **1B** is a preferred embodiment of a dry sprinkler assembly **10**. Preferred embodiments of the sprinkler can be configured for an upright installation, a pendent installation or a horizontal installation. The sprinkler assembly generally includes an elongate tubular outer housing **12** having a first end **14** and a second end **16** opposite the first end **14**. Within the tubular housing **12**, an internal conduit **18** extends from the first end **14** to the second end **16** along a central longitudinal sprinkler axis X-X. The first end **14** of the housing **12** defines a fluid intake end **10a** of the sprinkler assembly **10** having an inlet opening **20** and an internal sealing surface **22** proximate the inlet opening **20**. The second end **16** of the housing **12** defines a fluid discharge end **10b** of the sprinkler assembly **10** having an outlet opening **24**. Installed, the first end **14** of the sprinkler assembly **10** can be coupled to a fluid supply pipe of a sprinkler system with the central longitudinal sprinkler axis X-X in, for example, a preferably horizontal orientation parallel to the floor or ceiling for fluid discharge from the outlet opening **24** directed horizontally in the direction of the sprinkler axis X-X toward a fluid deflection member **30** affixed to the housing **12**. Preferred embodiments of the fluid deflection member **30** can direct the flow of fluid outwardly and downwardly, with some of the fluid lifted to project the fluid across a room, for example, and some of the fluid directed laterally downward to provide wall wetting.
- (8) Generally, fluid flow through the sprinkler assembly **10** is controlled by a fluid control assembly **100** disposed within the internal conduit of the outer housing **12** for displacement from a first position in fluid tight contact with the sealing surface **22** to a second position out of contact with the sealing surface **22** to permit the flow of firefighting fluid therethrough. The sprinkler **10** is a preferably an automatic sprinkler in which fluid flow through the sprinkler is regulated by a thermally responsive trigger assembly **39** and the preferred internal fluid control assembly **100** disposed within the housing **12**. The trigger **39** defines an unactuated state of the sprinkler assembly **10** in which the trigger **39** supports the internal fluid control assembly **100** within the housing **12** into contact with the internal sealing surface **22** to form a fluid tight seal with the internal sealing surface **22** and seal the rest of the sprinkler assembly from the supply pipe. Upon thermal operation of the trigger **39** in response to a level of heat indicative of a fire, an actuated state of the sprinkler assembly **10** is defined in which support of the fluid control assembly **100** has been removed which permits the internal fluid control assembly **100** to displace, and more preferably axially translate, out of contact with the internal sealing surface **22** under the fluid pressure in the fluid supply pipe of the system and/or an internal spring (not shown) that biases the fluid control assembly out of contact with the internal sealing surface **22**. Firefighting fluid delivered to the intake end **10a** of the sprinkler assembly flows through the internal conduit **18** and the internal fluid control assembly **100** and is discharged out of the outlet opening **24** of the housing **12** along a fluid flow path for effective fluid distribution fire protection by the fluid

deflection member **30** affixed to the housing **12** preferably at a fixed distance from the outlet opening **24** which defines a frame window therebetween.

(9) The fluid control assembly **100** includes an ejectable member that is translated out of the internal conduit **18** of the housing, ejected out the outlet opening **24** and displaced out of the fluid flow path between the outlet opening **24** and the fluid deflection member **30**. In the preferred sprinkler assembly **10**, a preferred structural and dynamic relationship between the ejectable member and the housing ensure proper guided and complete ejection and displacement of the ejectable member out of the fluid discharge fluid flow path. Generally, the ejectable member preferably defines a mechanical interface with other structures of the sprinkler assembly such as, for example, the housing which facilitates ejection of the ejectable member through the housing outlet opening and out of the preferred unencumbered fluid flow path upon thermal actuation of the sprinkler. More specifically, with the fluid control assembly **100** out of contact with the internal sealing surface **22**, preferred embodiments of the mechanical interface include a surface contact between the ejectable member of the fluid control assembly **100** and an internal or external surface of the housing **12** to guide the ejectable member out of the housing **12** and pivot the member out of the frame window and clear of fluid flow path. Alternatively, the sprinkler assembly **10** can include an intermediate member between the ejectable member and the housing such as, for example, a biasing member or spring to pivot the member out of the frame window and clear of fluid flow path. The member is ejected into the frame window with the member initially coaxially aligned with the central sprinkler axis and then skewed with respect to the central longitudinal sprinkler axis upon the member contacting the housing surface and more preferably an internal contact surface of the housing. Moreover, the preferred structural and dynamic relationship between the ejectable member and the housing **12** define a spatial and temporal coordination between the axial translation of the ejectable member and its pivot out of the fluid flow path by axially guiding the ejectable member and inhibiting or otherwise preventing its angular rotation about the central longitudinal axis X-X.

(10) In preferred embodiments of the sprinkler assembly **10**, the fluid deflection member **30** is located at a fixed distance from the outlet opening **24** and in the housing **12**, the second end **16** preferably includes a terminal portion **28** axially spaced downstream from the outlet opening **24** to which the fluid deflection member **30** can be affixed. The terminal portion **28** preferably defines the preferred frame window between the terminal portion **28** and the outlet opening **24**. The terminal portion **28** preferably, at least partially, circumscribes the central longitudinal sprinkler axis X-X, and more preferably, completely circumscribes the sprinkler axis X-X. Alternatively or additionally, in order to locate the deflector, the sprinkler housing **12** preferably includes a pair of frame arms **27a**, **27b** that are diametrically opposed about the outlet opening **24** extending axially away therefrom between the outlet opening **24** and the preferred terminal portion **28** toward the deflection member **30**. In an alternative embodiment, the frame arms **27a**, **27b** can converge toward the central longitudinal axis X-X and form a coaxially aligned fluid deflection boss, for example as seen in U.S. Pat. No. 8,636,075, to which the fluid deflection member **30** can be affixed. In such an embodiment, the deflection member **30** can include or define a central portion that, together with the deflection boss, presents an abutment to the fluid discharge from the outlet opening **24** to redirect and spread the discharged fluid from its center to fan the fluid radially outwardly to provide, for example, an effective horizontal fluid distribution.

(11) In preferred embodiments of the sprinkler assembly **10**, a fluid distribution with an unencumbered fluid flow path is provided. More particularly, embodiments of the sprinkler housing **12**, fluid control assembly **100** and fluid deflecting member **30** provide for the preferred unencumbered fluid flow path that extends from the outlet opening **24** to the fluid deflection member **30** and more preferably extends from the fluid control assembly **100** to the fluid deflection member **30**. For preferred embodiments of the sprinkler assembly **10**, the preferred fluid control assembly **100** includes the preferred ejectable member that clears the frame window between the

outlet opening and the terminal portion of the housing **12** upon sprinkler actuation to define the preferred unencumbered fluid flow path along the central longitudinal sprinkler axis from at least the outlet opening **24** to the terminal portion **28** and preferably through the terminal portion **28**. Preferred embodiments of the fluid deflection member **30** affixed to the housing **12** has one or more fluid flow surfaces that extends the preferred unencumbered fluid flow path from the terminal portion **28** of the housing to the fluid flow surface of the fluid deflection member **30**. For a fluid column discharged from the outlet opening **24**, the fluid column is acted on at its outer surface or periphery by the fluid deflection member **30** to direct the fluid stream in a desired manner to produce the fluid distribution for effective fire protection. Accordingly, preferred embodiment of the sprinkler assembly **10** provide that the unencumbered fluid flow path extends through the fluid deflection member **30**.

(12) In preferred embodiments of the sprinkler housing **12**, the terminal portion **28** preferably forms an annular member centered and circumscribed about the central longitudinal sprinkler axis. In preferred embodiments of the housing **12** having the pair of frame arms **27a**, **27b**, the arms terminate at the terminal portion **28** and more preferably at the annular formation **28**. In the actuated state of the preferred sprinkler assembly **10**, the preferred unencumbered fluid flow path extends from the outlet opening **24** to the annular formation **28** and more preferably through the annular formation **28**. The fluid deflection member **30** is preferably affixed to the outer housing **12** and more preferably affixed to the terminal portion **28** to locate the fluid deflection member **30** at the preferred fixed distance from the outlet opening **24**. Preferred embodiments of the fluid deflection member **30** include one or more fluid flow surfaces radially spaced from the central longitudinal axis for acting on the periphery of the discharge column from the outlet **24**. Accordingly, the fluid deflection member **30** preferably extends the preferred unencumbered fluid flow path along the central longitudinal sprinkler axis from the terminal portion **28** to the fluid flow surface.

(13) Preferred embodiments of the fluid deflection member **30** include at least one tab **32** that is radially spaced from the central longitudinal sprinkler axis X-X and which includes the preferred fluid flow surfaces, edges and/or slots to define preferred geometries for acting on the periphery of the discharge column. With specific reference to FIGS. **1A** and **1B**, preferred embodiments of the fluid deflection member **30** more preferably includes at least two tabs, such as for example, a preferred first tab **32a** and a preferred second tab **32b**. The first and second tabs **32a**, **32b** are opposed from one another about a first plane **P1** defined by the central longitudinal sprinkler axis X-X and a lateral axis Y-Y extending perpendicular to the central longitudinal sprinkler axis X-X to define the preferred unencumbered fluid flow path extending from the outlet opening **24**, and more preferably from the fluid control assembly along the central longitudinal sprinkler axis X-X to the tabs **32a**, **32b**. The preferred fluid deflection member **30** can be configured similarly to the flow-shaping member as shown and described in any one of U.S. Pat. Nos. 8,662,190; 8,151,462 and 7,712,218. Accordingly, preferred embodiments of the sprinkler assembly can provide for the preferred unencumbered fluid flow path to extend from the outlet opening **24**, and more preferably from the fluid control assembly **100**, through the fluid deflection member **30** along the central longitudinal sprinkler axis X-X to the tabs **32a**, **32b**.

(14) As seen in FIG. **1A** and FIG. **2**, each of the tabs **32a**, **32b** are preferably angled with respect to the sprinkler axis X-X to present inwardly facing fluid flow surfaces to the outlet **24**. With particular reference to FIG. **2**, each of the preferred first and second tabs **32a**, **32b** have a leading edge **34a**, **34b** and a trailing edge **36a**, **36b** with the fluid flow surfaces **38a**, **38b** extending therebetween. Each of the first and second tabs **32a**, **32b** are angled and more preferably skewed with respect to the central longitudinal sprinkler axis X-X so that the leading edge **34a**, **34b** is radially inward of the trailing edge **36a**, **36b**. The angle of the tabs **32a**, **32b** preferably taper the unencumbered fluid flow path. Each of the tabs **32a**, **32b** define a preferred included angle with the central longitudinal sprinkler axis X-X that ranges from thirty degrees to sixty degrees)(30°-60°.

The included angles of the tabs can be the same or different. In one preferred embodiment, the first tab **32a** defines a preferred included angle ranging from 35°-40° and is more preferably 37 degrees. The second tab **32b** defines a different included angle ranging from 30°-50° and more preferably being any one of 33° and 48° with the central longitudinal sprinkler axis X-X.

(15) The tabs **32a**, **32b** and their edges each define a preferably polygon-shaped geometry with features that can be similar to one another. For example, each of the preferred tabs **32a**, **32b**, can have parallel lateral edges that extend perpendicularly between the leading and trailing edges. The spacing between the lateral edges define the width of the tabs **32a**, **32b** with the length of the lateral edges defining the length of the tabs **32a**, **32b**. The widths of the tabs **32a**, **32b** may similarly or variably range between 0.300 inch 3.000 inches and lengths of the tabs **32a**, **32b** can similarly or variably range between 0.200 to 1.300 inches. More preferably, the tabs **32a**, **32b** are geometrically configured differently. In the preferred embodiment of the fluid deflection member **30** of FIG. 2, the leading edge **34a** of the first tab **32a** preferably defines a width ranging between 0.5 inch to 0.66 inch with a plurality of spaced apart open-end slots **40**. Each of the open-end slots **40** initiate from and extend from the leading edge **34a** in a direction perpendicular to the leading edge **34a** to terminate at a terminal end of the slot **40**. The plurality of open-end slots **40** preferably includes a central slot with two lateral slots disposed equidistantly about the central slot. The lateral slots each have a slot length that is preferably greater than the slot length of the central slot.

(16) In a preferred fluid deflection member **30**, the leading edge **34b** of the second tab **32b** preferably defines a width smaller than the leading edge **34a** of the first tab **32a** with a central linear edge portion and two lateral linear edge portions disposed about the central portion. The leading edge **34b** of the second tab **32b** is preferably configured such that the central linear edge portion is closer to the leading edge **34a** of the first tab **32a** than the two lateral linear edge portions of the second leading edge **34b**. The second tab **32b** also preferably includes a central closed formed slot **42** extending in a direction perpendicular to the leading edge. Moreover, in another preferred aspect, the trailing edge **36b** of the second tab **32b** includes a pair of open-ended slots **44** disposed about the central linear edge portion at the leading edge **34b** and the central slot **42**. The open-ended slots **44** initiate from the trailing edge **36b** toward the leading edge **34b** of the second tab **32b**.

(17) The one or more tabs **32** can be integrally formed with or affixed, directly or indirectly, to the housing **12**. For example, the tabs **32a**, **32b** can be affixed to or integrally formed with the preferred annular boss **28**. More preferably, the tabs **32a**, **32b** are formed with an extend from an annular base **46** which is preferably affixed internally to the annular boss **28** of the housing **12**.

Accordingly, the annular base **46** of the fluid deflection member **30** is dimensioned to be centered within the annular boss **28** and moreover is preferably dimensioned to define and maintain the unencumbered fluid flow path of the sprinkler assembly **10**. With reference to FIGS. 1A and 1B, the fluid deflection member **30** is oriented with respect to the frame arms **27a**, **27b**. In particular, the tabs **32a**, **32b** are preferably located so as be perpendicular to the frame arms **27a**, **27b**. The frame arms **27a**, **27b** are preferably disposed in and aligned with one another along a second plane P2 that is defined by the central longitudinal axis X-X and a vertical axis Z-Z which extends perpendicular to the first plane P1. Accordingly, the fluid deflection member **30** is oriented such that the first and second planes P1, P2 are perpendicular to one another with their intersection aligned along the central longitudinal sprinkler axis X-X. In the preferred geometry of the fluid deflection member **30**, the deflection member **30** is symmetrically bisected by the second plane P2. In the preferred installation of the sprinkler assembly **10**, the first plane P1 is oriented parallel to the floor or ceiling with the first tab **32a** above the second tab **32b** and the frame arms **27** vertically aligned with one another and the second plane P2 disposed perpendicular to the floor or ceiling.

(18) The housing **12** and the fluid control assembly **100** define and maintain the preferred unencumbered fluid flow path of the preferred assembly **10** by keeping operational components clear of the fluid flow path upon sprinkler operation. Referring again to FIGS. 1A and 1B, a

preferred embodiment of the fluid control assembly **100** includes a seal subassembly **102** and a fluid flow tube **104** which forms a discharge orifice end **106** opposite the seal subassembly **102**. Abutting the discharge orifice end **106** is a support subassembly **110** which forms the preferred ejectable member of the fluid control assembly **100**. Generally, the ejectable support subassembly **110** includes a post member **112** with a projection member **114** affixed to the post member **112** that extends radially outward from the post member **112**. Within the housing **12** is an internal contact surface or shelf **26** formed proximate the outlet opening **24**. Adjacent the contact shelf **26**, the internal surface of the housing **12** preferably includes a formed axially extending channel **62** proximate the outlet opening **24** contiguous with the internal shelf **26**. The projection member **114** is received within the channel **62** to axially and rotationally guide the support subassembly **110** and the rest of the fluid control assembly **100** toward the internal contact shelf **26** upon thermal actuation of the sprinkler assembly. The post member **112** is ejected out of the outlet opening **24** to bring the projection member **114** in contact with the internal shelf **26** so as to impart a rotation on the support subassembly **110** and pivot the support subassembly **110** out of the fluid flow path from the outlet opening **24** to the fluid deflection member **30**.

(19) Shown in FIGS. **3A** and **3B** are detailed partial cross-sectional views of the fluid discharge end **10b** of the sprinkler assembly **10** of FIGS. **1**, **1A** and **1B** showing a preferred structural and dynamic relationship defined by a preferred mechanical interface between the support subassembly **110** and the internal surface of the housing **12**. Although the tubular housing **12** can be formed as a single unitary structure, the tubular housing is more preferably formed by the interconnection of two or more tubular housing components. For example, the housing **12** preferably includes an externally threaded body **50** forming the fluid discharge end **10b**, another externally threaded tubular component **52** forming the fluid intake end **10a**, with an intermediate internally threaded tubular component **54** interconnecting the fluid inlet and discharge end components **50**, **52**. The components of the housing **12** can be joined by alternate means or configurations provided the assembly provides for the internal conduit **18** and fluid intake and discharge ends **10a**, **10b** as described herein. The fluid discharge end **10b** of the housing **12** preferably includes the preferred externally threaded body **50**, as shown in FIGS. **3A** and **3B**, with an internal surface **60** in which the preferred axially extending channel **62** is formed with the preferred internal contact shelf **26** between the channel **62** and the outlet opening **24**. The channel **62** is dimensioned and configured to accommodate the projection member **114** of the support subassembly **110** and guide its axial translation toward the internal contact shelf **26** and otherwise constrain angular rotation of the support subassembly **110** about the sprinkler axis X-X. In an alternate embodiment of the sprinkler **10**, the internal surface **60** can include the affixed projection member and the support assembly **110** can include the channel formation with an appropriately located contact shelf or surface. In an inverse cooperative relationship, the projection member and channel would axially guide the support subassembly **110** and its shelf formation toward the projection member and resist angular rotation of the support subassembly **110** about the sprinkler axis X-X for its ejection and pivot out of the fluid flow path in a manner as previously described.

(20) In the preferred embodiments shown, the recessed channel region **62** is defined by a depth DP measured in the radial direction preferably from the central axis X-X, a width WD1 measured perpendicular to the radial direction between a pair of channel sidewalls **64** and its axial length LD which is preferably 3.5 to 4 times greater than the width WD1. The width WD1 is sufficiently broad to permit axial translation of the projection member **114** within the channel **62** to contact the internal contact surface **26** and sufficiently narrow to limit or otherwise inhibit and more preferably prevent rotation of the support subassembly **110** about the sprinkler axis X-X and the relative rotation between the support subassembly **110** and the outer housing **12**. The channel **62** is preferably located so as to be centered between the frame arms **27a**, **27b** to locate the pivot for the support subassembly **110** that is centered between the frame arms **27a**, **27b**. The width WD1 of the channel **62** is greater than a width WD2 of the projection member **114** and preferably 10-30%

greater than the width of the projection member **114** and more preferably 10-15% greater than the width WD2 of the projection member **114**. In a preferred embodiment in which the channel width WD1 is preferably no more than 1.25 times the width WD2 of the projection member **114** and more preferably 1.2 to 1.15 times the width WD2 of the projection member **114**. The depth DP of the channel **62** preferably increases in the axial direction toward the internal shelf **26**. In another preferred aspect, the preferred channel **62** defines one or more dimensional relationships with other features of the externally threaded body **50**, for example, the channel width and length define preferred respective ratios with the diameter DIA of the outlet opening **24**. For example, a preferred outlet diameter-to-channel width ratio (DIA:WD1) preferably ranges from 3.5:1 to 4:1 and is preferably 3.75:1. A preferred channel length-to-outlet diameter ratio (LD:DIA) preferably ranges from 1:1 to 1.1:1. In a preferred embodiment, the outlet diameter DIA is 0.75 inch.

(21) Shown in FIGS. **4** and **4A** are various views of a preferred support subassembly **110** for use in the flow control assembly **100**. The post member **112** preferably includes a cylindrical body portion **120** having a first diameter D1 and a cylindrical head portion **122** of a second diameter D2 smaller than the first diameter with a neck portion **124** formed between the body and head portions **120**, **122** having a third diameter D3 greater than the second diameter D2. Alternatively, the diameters post member **112** can be equal to one another or vary from one another in any manner provided the post member **112** provides for the support and ejection of the support assembly **110** in a manner as described herein. The body portion **120** is preferably a right circular cylinder but can define alternate geometries. For example, a preferred embodiment of the body portion can include a chamfered portion **126** as shown in FIG. **4**, which can offset the center of gravity of the post member from the sprinkler axis X-X to facilitate the pivoted rotation of the subassembly **110**. More preferably, the chamfer is diametrically aligned opposite the projection **114** of the subassembly. The support subassembly **110** remains generally coaxially centered with respect to the sprinkler axis X-X from its position in the unactuated state of the sprinkler assembly **10** through the axial displacement of the support subassembly **110** in the actuated state of the sprinkler assembly **10** until the projection member **114** contacts the internal contact surface **26**. In a preferred aspect of the structural and dynamic relationship between the housing **12** and the support subassembly **110**, the diameter D1 of the body portion **120** defines a maximum external diameter of the post member **112** and is smaller than the internal diameter DIA of the outlet opening **24** to define an internal diameter-to-maximum external diameter ratio (DIA:D1) that ranges from 1.1:1 to 1:1.

(22) In the support subassembly **110**, the projection member **114** preferably extends radially from the post member **112** and more preferably from the neck portion **124**. As shown, the projection member **114** is preferably a separate component disposed and secured about the head and neck portions **122**, **124** of the post member **112**. The preferred projection member **114** includes an arcuate portion **116a** that at least partially circumscribes and more preferably completely circumscribes the neck portion **124** of the post member **112** and a rectilinear portion **116b** extending radially from the arcuate portion. The support subassembly **110** preferably includes a pip cap **130** centered within the cylindrical body **120** to support the thermally responsive trigger **39** in the unactuated state of the sprinkler assembly. The support subassembly **110** is seated against the thermally responsive trigger **39** to locate the fluid flow assembly **100** within the housing **12** such that the projection member **114** is within the channel **62** and axially spaced from the internal contact surface **26**. In the unactuated state of the assembly, the seal subassembly **102** forms a fluid-tight sealed engagement with the internal sealing surface **22**. Together, the post member **112** and the pip cap **130** preferably substantially fill the outlet opening **24** substantially concealing the internal conduit **18** of the housing **12**. In the actuated state of the sprinkler assembly **10** upon thermal actuation of the trigger **39** and ejection of the support subassembly **110**, the remainder of the fluid control assembly **100** is axially translated in which the seal subassembly **102** is spaced from the sealing surface **22**.

(23) In the unactuated state of the sprinkler assembly **10**, the thermally responsive trigger **39** is

seated preferably at a fixed distance from the outlet opening **24**, as shown in FIGS. **1** and **1A**, to transfer a compressive load to the fluid control assembly **100** and form the sealed engagement at the internal sealing surface **22**. In the preferred embodiment, the trigger **39** comprises a frangible glass bulb having one end preferably seated at or proximate the frame boss **28** under load from one or more screw members **41** threadedly engaged with the frame boss **28**. Alternatively, the trigger **39** can be configured as a soldered mechanical assembly seated proximate the frame boss **28**. The trigger **39** has a nominal operating temperature and thermal sensitivity to define the thermal responsiveness of the sprinkler at which the sprinkler actuates in response to a fire. In preferred embodiments of the sprinkler assembly **10**, the trigger **39** has a preferred nominal operating temperature rating that ranges between 125° F. to 225° F. (52° C.-107° C.) and more preferably is any one of: 155° F. (68° C.); 175° F. (79° C.) or 200° F. (93° C.). The thermal sensitivity of a trigger assembly and sprinkler is measured or characterized by Response Time Index ("RTI"), measured in units of $(\text{ft} \cdot \text{Math.s})^{\text{sup.}1/2} [(\text{m} \cdot \text{Math.s})^{\text{sup.}1/2}]$. An RTI of 145-635 $(\text{ft} \cdot \text{Math.s})^{\text{sup.}1/2}$ [80 $(\text{m} \cdot \text{Math.s})^{\text{sup.}1/2}$ to 350 $(\text{m} \cdot \text{Math.s})^{\text{sup.}1/2}$] defines a "Standard Response Sprinkler" and an RTI equal to or less than 90 $(\text{ft} \cdot \text{Math.s})^{\text{sup.}1/2}$ [50 $(\text{m} \cdot \text{Math.s})^{\text{sup.}1/2}$] defines a "Quick Response Sprinkler." Preferred embodiment of the sprinkler assembly are configured as a quick response sprinkler.

(24) In the preferred embodiment of the sprinkler assembly **10** shown in FIGS. **1**, **1A-1B** and **3A**, the glass bulb trigger **39** is seated against a preferred yoke member **200** to align the glass bulb trigger **39** along the central sprinkler axis X-X and the preferred unencumbered fluid flow path. Generally, the preferred yoke member **200** is configured in a manner similar to the yoke shown and described in U.S. Pat. No. 10,238,903. The preferred yoke member **200** includes a crossbar portion **202** with a central region **204** for seating the end of the glass bulb trigger **39** opposite the support subassembly **110**. The crossbar portion **202** also include two end regions **206a**, **206b** disposed about the central region **204** that are each subject to a load force to axially load the glass bulb **39** and fluid control assembly **100**. In a preferred embodiment, the crossbar portion **202** is preferably formed with the central region **204** located axially further away from the outlet opening **24** than the two end regions **206a**, **206b**. The crossbar portion **202** is preferably aligned with the frame arms **27a**, **27b** in the vertically extending plane P2. The assembly includes two load screws **41** threadedly engaged with the annular boss **28** to apply a compressive force respectively to the end regions **206a**, **206b** of the crossbar portions **202**. The yoke member **200** preferably includes an extension member **208** extending between the two end regions **206a**, **206b** of the yoke member **200**. The extension member **208** preferably extends from the crossbar portion **202** so as to be skewed with respect to the central longitudinal sprinkler axis X-X as shown in FIG. **1B**. The extension member **208** can define a center of gravity of the yoke member **200** that is off-set from the central longitudinal sprinkler axis X-X to facilitate rotation and clearance of the yoke member **200** out of the fluid flow path upon sprinkler upon sprinkler actuation.

(25) In a preferred horizontal installation of the sprinkler assembly **10**, upon sprinkler thermal actuation in which the trigger **39** ruptures, the preferred support subassembly **110** is ejected horizontally parallel to the floor and the seal subassembly **102** and fluid flow tube **104** translate horizontally toward the outlet opening **24**. When the projection member **114** contacts the internal contact surface **26**, the support assembly **110** pivots between the frame arms **27a**, **27b** about an axis parallel to Z-Z axis and clear of any sprinkler structure to avoid any lodgment of the support subassembly **110**. With the support subassembly **110** ejected clear of the sprinkler assembly **10**, the inlet opening **20** and the discharge orifice are fully open and the preferred unencumbered fluid flow path are clear for flow of firefighting fluid therethrough to impact the fluid deflection member **30**.

(26) The remaining components of the preferred fluid control assembly **100**, including the seal assembly **102** and the fluid flow tube **104** can each be configured and assembled using multiple components. For example, as shown in FIG. **1A**, the seal assembly **102** preferably includes a spring disc **101** affixed about a base **103** having an array of legs **103a** extending therefrom. In the

unactuated state of the sprinkler assembly, the spring disc **101** forms the fluid tight sealed contact with the internal seal surface **22** of the housing. The seal assembly **102** can be configured as any one of the embodiments of “spring support assembly” shown and described in the dry sprinkler assembly of U.S. Pat. No. 8,636,075. The fluid flow tube can be a single tube or made from multiple tubes. The support subassembly **110** is preferably received within the discharge orifice **106** in an abutting engagement. The seal assembly can be biased in a direction away from the sealing surface **22** by an internal spring member disposed about the first tubular member (not shown). Alternatively, the fluid control assembly **100** can include one or more internal link and plug components shown in U.S. Pat. No. 7,921,928.

(27) In the actuated and open state of the sprinkler assembly **10**, the displacement and translation of the preferred fluid control assembly **100** ejects the supporting subassembly **110** and locates the discharge orifice **106** within the body **50** at the fluid discharge end **10b** of the housing **12** proximate the outlet opening **24**. Fluid flowing through the inlet opening **20** flows at a preferred operating pressure, through the fluid flow tube **104**, out the discharge orifice **106** and the outlet opening **24** along the preferred unencumbered fluid flow path to define the fluid discharge column that is acted upon by the axially spaced fluid deflection member **30**. The discharge orifice is preferably configured and dimensioned to define the desired discharge characteristics of the sprinkler. Accordingly, the discharge orifice **106** can be quantified by a preferred nominal K-factor. The discharge or flow characteristics from the sprinkler body is defined by the internal geometry of the sprinkler including its internal passageway, inlet and outlet (the orifice). As is known in the art, the K-factor of a sprinkler is defined as $K=Q/P^{.5}$, where Q represents the flow rate (in gallons/min GPM) of water from the outlet of the internal passage through the sprinkler body and P represents the pressure (in pounds per square inch (psi.)) of water or firefighting fluid fed into the inlet end of the internal passageway through the sprinkler body. Generally, the discharge characteristics of the sprinkler body define a preferred nominal K-factor in a range of 4 [GPM/(psi)^{.5}] to 50 [GPM/(psi)^{.5}]. Preferred embodiments of the sprinkler body define a nominal K-factor which preferably ranges from a nominal 4.0 [GPM/(psi)^{.5}] to 14.0 [GPM/(psi)^{.5}]. More preferably, the sprinkler body defines a K-factor of any one of 4.0 [GPM/(psi)^{.5}]; 4.2 [GPM/(psi)^{.5}] or 4.4 [GPM/(psi)^{.5}]. Alternatively, the sprinkler body can define K-factors smaller or larger than the preferred range depending upon the application.

(28) While the present invention has been disclosed with reference to certain embodiments, numerous modifications, alterations, and changes to the described embodiments are possible without departing from the sphere and scope of the present invention, as defined in the appended claims. Accordingly, it is intended that the present invention not be limited to the described embodiments, but that it has the full scope defined by the language of the following claims, and equivalents thereof.

Claims

1. A dry sprinkler assembly comprising: a tubular outer housing having a first end and a second end opposite the first end with an internal conduit extending from the first end to the second end along a central longitudinal sprinkler axis, the first end defining a fluid intake end of the sprinkler assembly having an inlet opening and an internal sealing surface proximate the inlet opening, the second end defining a fluid discharge end of the housing having an outlet opening, the second end of the housing including a terminal portion of the housing circumscribed about the central longitudinal sprinkler axis, the terminal portion of the housing being axially spaced downstream from the outlet opening; and a fluid control assembly disposed coaxially within the internal conduit of the outer housing for displacement from being in fluid tight contact with the sealing surface to being out of contact with the sealing surface to provide for an unencumbered fluid flow path

extending along the central longitudinal sprinkler axis from the fluid control assembly through the terminal portion; and a fluid deflection member affixed to the terminal portion of the housing, the fluid deflection member having a fluid flow surface to maintain the unencumbered fluid flow path along the central longitudinal sprinkler axis between the fluid control assembly and through the fluid deflection member when the fluid control assembly is out of contact with the sealing surface.

2. The dry sprinkler assembly of claim 1, wherein the fluid deflection member has at least one tab radially spaced from the central longitudinal sprinkler axis at a fixed distance from the outlet opening to define the fluid flow surface and maintain the unencumbered fluid flow path along the central longitudinal sprinkler axis extending between the fluid control assembly and the at least one tab.
3. The dry sprinkler assembly of claim 2, wherein the terminal portion is an annular formation circumscribed about the central longitudinal sprinkler axis.
4. The dry sprinkler assembly of claim 2, wherein the fluid deflection member includes an annular base with the at least one tab extending from the annular base in the direction of the unencumbered fluid flow path, the annular base being affixed to the terminal portion of the housing and centered about the central longitudinal sprinkler axis such that the unencumbered fluid flow path extends along the central longitudinal sprinkler axis between the fluid control assembly and the at least one tab and through the annular base.
5. The dry sprinkler assembly of claim 2, wherein the at least one tab has a trailing edge and a leading edge with the fluid flow surface therebetween, the at least one tab being angled with respect to the central longitudinal sprinkler axis so that the leading edge is radially inward of the trailing edge.
6. The dry sprinkler assembly of claim 5, wherein the at least one tab includes a first tab and a second tab opposed from one another about a plane defined by the central longitudinal sprinkler axis and a lateral axis extending perpendicular to the central longitudinal sprinkler axis.
7. The dry sprinkler assembly of claim 6, wherein the plane extending between the first and second tabs defines a first plane and the second end of the tubular outer housing includes a pair of frame arms diametrically opposed about the outlet opening and disposed in a second plane that is perpendicular to the first plane, the pair of frame arms extending in a direction from the outlet opening toward the terminal portion.
8. The dry sprinkler assembly of claim 7, wherein each of the first and second tabs has a trailing edge and a leading edge with the fluid flow surface therebetween, each of the first and second tabs being angled with respect to the central longitudinal sprinkler axis so that the leading edge is radially inward of the trailing edge.
9. The dry sprinkler assembly of claim 8, wherein the leading edge of the first tab and the leading edge of the second tab are spaced apart so as to taper the unencumbered fluid flow path.
10. The dry sprinkler assembly of claim 8, wherein the leading edge of the first tab defines a plurality of spaced apart open end slots extending from the leading edge in a direction perpendicular to the leading edge.
11. The dry sprinkler assembly of claim 10, wherein the plurality of open end slots includes a central slot defining a slot length and two lateral slots disposed equidistantly about the central slot, the lateral slots each having a slot length greater than the slot length of the central slot.
12. The dry sprinkler assembly of claim 8, wherein the leading edge of the second tab defines a central linear edge portion and two lateral linear edge portions disposed about the central linear edge portion, the central linear edge portion being closer to the first plane than the two lateral linear edge portions.
13. The dry sprinkler assembly of claim 12, wherein the second tab includes a closed formed slot extending in a direction perpendicular to the leading edge.
14. The dry sprinkler assembly of claim 12, wherein the trailing edge of the second tab includes a pair of open-ended slots disposed about the central linear edge portion and extending in a direction

toward the leading edge of the second tab.

15. The dry sprinkler assembly of claim 1, wherein the fluid control assembly includes an ejectable member and the tubular outer housing defining a mechanical interface such that when the fluid control assembly is out of contact with the internal sealing surface, the ejectable member is ejected through the housing and out of the unencumbered fluid flow path.

16. The dry sprinkler assembly of claim 15, wherein the fluid discharge end of the housing includes an internal shelf proximate the outlet opening and the fluid control assembly includes: a seal subassembly; and a fluid flow tube abutting the seal subassembly; wherein the ejectable member comprises a support subassembly abutting the fluid flow tube, the support subassembly including a post member defining an axial length of the support subassembly with a projection member extending radially from the post member, wherein, in an unactuated state of the sprinkler assembly, the post member and projection member of the support subassembly are located within the fluid discharge end of the housing with the projection member axially spaced from the internal shelf; and wherein, in an actuated state of the sprinkler assembly, the support subassembly defines the ejectable member ejected through the outlet opening such that the projection member contacts the internal shelf so as to form the mechanical interface and rotate the support subassembly out of the unencumbered fluid flow path.

17. The dry sprinkler assembly of claim 16, wherein the support subassembly engages a thermally responsive trigger in the unactuated state of the sprinkler assembly.

18. The dry sprinkler assembly of claim 17, wherein the thermally responsive trigger is a glass bulb disposed along the central longitudinal sprinkler axis, the glass bulb having a first end engaging the support subassembly and a second end engaging a yoke member that includes a crossbar portion with a central region for seating the second end of the glass bulb and two end regions disposed about the central region that are each subject to a load force to axially load the glass bulb and fluid control assembly, the yoke member including an extension member extending between the two end regions of the crossbar portion to define a center of gravity that is off-set from the central longitudinal sprinkler axis.

19. A dry sprinkler assembly comprising: a tubular outer housing having a first end and a second end opposite the first end with an internal conduit extending from the first end to the second end, the first end defining a fluid intake end of the housing having an inlet opening and an internal sealing surface proximate the inlet opening, the second end defining a fluid discharge end of the housing having an outlet opening and a terminal portion spaced from the outlet opening and circumscribed about a sprinkler axis; a fluid control assembly disposed within the internal conduit of the tubular outer housing, a thermally responsive trigger located proximate the outlet opening, the thermally responsive trigger supporting the fluid control assembly to define an unactuated state of the dry sprinkler assembly with the fluid control assembly in fluid tight contact with the internal sealing surface, the thermally responsive trigger having a thermal operation defining an actuated state of the dry sprinkler assembly to provide a displacement of the fluid control assembly and an unencumbered fluid flow path along the sprinkler axis from the fluid control assembly and through the terminal portion, and a fluid deflection member affixed to the terminal portion, wherein the fluid deflection member comprises a fluid flow surface to provide the unencumbered fluid flow path through the fluid deflection member during the actuated state of the sprinkler assembly.

20. The dry sprinkler assembly of claim 19, wherein the fluid deflection member has at least one tab radially spaced from the sprinkler axis at a fixed distance from the outlet opening to define the fluid flow surface and maintain the unencumbered fluid flow path along the sprinkler axis.

21. The dry sprinkler assembly of claim 20, wherein the at least one tab includes a first tab and a second tab opposed from one another about a plane defined by the sprinkler axis and a lateral axis extending perpendicular to the sprinkler axis, and wherein each of the first and second tabs has a trailing edge and a leading edge with the fluid flow surface therebetween, each of the first and second tabs being angled with respect to the sprinkler axis so that the leading edge is radially

inward of the trailing edge.

22. The dry sprinkler assembly of claim 21, wherein the leading edge of the first tab and the leading edge of the second tab are spaced apart so as to taper the unencumbered fluid flow path.

23. The dry sprinkler assembly of claim 19, wherein the sprinkler axis comprises a central longitudinal sprinkler axis extending through the inlet opening and the outlet opening.

24. The dry sprinkler assembly of claim 19, wherein the terminal portion comprises an annular formation completely circumscribed about the sprinkler axis.

25. A dry sprinkler assembly comprising: a tubular housing having an internal conduit defined between a first end and a second end; a fluid inlet end component connected in fluid communication with the first end of the tubular housing, the fluid inlet end component having an inlet opening, an internal sealing surface proximate the inlet opening, and a seal assembly in fluid tight contact with the internal sealing surface; a fluid discharge end component connected in fluid communication with the second end of the tubular housing, the fluid discharge end component having an outlet opening and a terminal portion spaced from the outlet opening and circumscribed about a sprinkler axis; a thermally responsive trigger proximate the outlet opening, the thermally responsive trigger defining an unactuated state of the dry sprinkler assembly, the thermally responsive trigger having a thermal operation defining an actuated state of the dry sprinkler assembly to displace the seal assembly from the internal sealing surface; and a fluid deflection member affixed to the terminal portion, the fluid deflection member having at least one tab with a fluid flow surface radially spaced from the sprinkler axis at a fixed distance from the outlet opening to provide an unencumbered fluid flow path through the fluid deflection member during the actuated state of the sprinkler assembly.

26. The dry sprinkler assembly of claim 25, wherein the at least one tab includes a first tab and a second tab opposed from one another and disposed about the sprinkler axis, and wherein each of the first and second tabs has a trailing edge and a leading edge with the fluid flow surface therebetween, each of the first and second tabs being angled with respect to the sprinkler axis so that the leading edge is radially inward of the trailing edge.

27. The dry sprinkler assembly of claim 25, wherein the sprinkler axis comprises a central longitudinal sprinkler axis extending through the inlet opening and the outlet opening.

28. The dry sprinkler assembly of claim 25, wherein the terminal portion comprises an annular formation completely circumscribed about the sprinkler axis.
